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The effects of an open-lung approach during one-lung ventilation on postoperative pulmonary complications and driving pressure: a descriptive multicenter national study.

Running head: OLA during one-lung ventilation

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Declaration of interest

None

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Presentation

None

Abstract

Background: Thoracic surgical procedures are associated with an increased risk of postoperative pulmonary complications (PPCs), which seem directly related to intraoperative driving pressure.

Objective: We conducted this study to describe the incidence of PPCs in patients in who an individualized open-lung approach was applied during one-lung ventilation.

Design: This was a prospective, multicenter and national descriptive study.

Setting: Thoracic surgery patients undergoing one-lung ventilation.

Patients: Eligible participants were consecutively included from October 1, 2016 to September 30, 2017

Interventions: An individualized open lung approach which consisted in an alveolar recruitment maneuver followed by a positive end-expiratory pressure adjusted to best respiratory system compliance was performed in all patients.

Main outcomes and measures: Pre- and intraoperative data were recorded; the primary outcome was a description of the incidence of PPCs in these patients during the first 7 postoperative days.

Results: A total of 690 patients were included. The patients were mainly male and half of them had a high-risk of PPCs (ARISCAT score exceeding 44); 11% of them developed a PPC within the first postoperative week. The mean open lung positive end-expiratory pressure was 8 ± 3 cmH_2O. When compared to pre-open lung approach values, open-lung approach significantly decreased the driving pressure (14 ± 4 vs. 11 ± 3 cmH_2O, $p < 0.001$) and increased dynamic compliance (30 ± 10 vs. 43 ± 15 ml/cmH_2O, $p < 0.001$).

Conclusions: The low incidence of PPCs in patients who received an open-lung approach one-lung ventilation compared to that reported for other thoracic surgery series, and the decrease in the driving pressure in these patients justifies a further randomized controlled trial to compare the open-lung approach to the standard protective strategy of low tidal volume and low positive end-expiratory pressure.

Clinical trial registration: NTC03184974.

Introduction

Thoracic surgery procedures are associated with an increased risk of postoperative pulmonary complications (PPCs) as a result of several factors associated with the patient, the surgery itself, or others, such as ventilation induced lung injury (VILI).[1,2] Although the incidence of moderate to severe PPCs after thoracic surgery is not accurately known, it ranges between 15% to 32% in studies with large sample size.[3-5] Lung protective ventilation seems to minimize PPCs by decreasing the main mechanisms of VILI: overdistension and atelectrauma.[6] Although the use of lower tidal volumes (VT) is associated with a reduction in PPCs during both two- and one-lung ventilation (OLV), it is not clear whether positive end-expiratory pressure (PEEP) with or without previously associated alveolar recruitment maneuver (RM) reduces the risk of PPCs.[7] Recent data suggest that the combination of low VT and sufficient PEEP reduces PPCs.[5,8] The open-lung approach (OLA) is a ventilator strategy involving low VT, RMs, and a decremental PEEP titration[9,10]. The RM aims to open the lungs and the PEEP is titrated to the level of best dynamic compliance (cdyn) at which the lungs are maintained open. Thus, the PEEP titration results in a more individualized PEEP adjustment.

A key factor associated with PPCs is the driving pressure (DP), i.e the pressure gradient needed to generate a given tidal volume calculated as plateau pressure (Pplat) minus PEEP. Our group has recently shown that an OLA reduces the DP in lung-healthy patients undergoing abdominal surgery,[11,12] and this concurs with several other studies that suggest similar results during OLV.[13,14] Intraoperatively, DP seems to be directly related to the occurrence of PPCs during two- and OLV.[15,16] Moreover, it has recently been shown in an experimental study that low DP is associated with reduced lung damage[17] .Therefore, it has been postulated that protective intraoperative mechanical ventilation should be aimed at reducing DP.[18]

The main objective of this present study was to describe the incidence of PPCs in patients who underwent an OLA procedure during OLV and to describe the effects of the OLA on DP. In addition, we attempt to describe the association between the individualized PEEP level after lung recruitment and the patient's preoperative or intraoperative variables.

Material and Methods

Study design

This prospective, descriptive, multicenter, national study was approved by the Institutional Review Board at each participating center and is registered at clinicaltrials.gov (NTC03184974). The written informed consent of each patient was obtained. This manuscript adheres to the STROBE guidelines[19]

Participants

A total of 14 Spanish teaching hospitals participated in the study. Eligible participants were consecutively included from October 1, 2016 to September 30, 2017. Eligible patients were aged 18 years or over and were submitted to general anesthesia which required OLV and in which the anesthesiologist responsible considered open lung PEEP (OL-PEEP) as the standard-of-care for intraoperative mechanical ventilation. There were no other specific exclusion criteria. The total number of eligible participants was retrospectively reported.

Study protocol

After 5 minutes breathing 100% oxygen, anesthesia was induced with fentanyl 5 μg/kg^{-1}, propofol 2.5 mg.kg^{-1}, and rocuronium 0.6 mg/kg^{-1}. Sevoflurane was administered to maintain a bispectral index between 40 and 50. Patients received fentanyl or a continuous infusion of remifentanil, and crystalloid solutions were continuously infused at a rate of 3 mL/kg^{-1}/h. The trachea was intubated with an appropriately sized left-side double-lumen tube; the tube position was confirmed by bronchoscopy in the supine and lateral positions.

The patient's lungs were ventilated during TLV and OLV in volume-controlled ventilation with square-wave flow. TV was set to 8 ml/kg^{-1} of the predicted body weight (PBW) during two-lung ventilation and to 5–6 mL/kg^{-1} for OLV to maintain a plateau pressure not exceeding 25 cmH$_2$O. Fraction of inspired oxygen (FiO$_2$) of 80%. The inspiratory-to-expiratory ratio was 1:2 with an end-inspiratory pause of 10%, and the frequency was adjusted to maintain the arterial carbon dioxide partial pressure (PaCO$_2$) between 4.6 and 8 kPa; all the patients had an initial PEEP level of 5–6 cmH$_2$O during two-lung ventilation.

The alveolar RM and PEEP titration trial were performed at the beginning of the OLV, (as described elsewhere[10]). The OLA, which involves the RM and PEEP titration trial, consists of four sequential steps: (1) First a stepwise alveolar recruitment maneuver (RM) is performed: the ventilatory mode is switched to pressure controlled ventilation (PCV) and adjusted to a driving pressure (DP) of 20 cmH$_2$O and an initial

PEEP of 5 cmH_2O. PEEP is then increased in 5 cmH_2O increments, each lasting 10 respirations until a PEEP titration value of 20 cmH_2O (plus 20 cmH_2O of DP, accounting for a total airway opening pressure of 40 mH_2O for 15 respirations). (2). The ventilation mode is then switched back to VCV with the same baseline settings but without modifying the PEEP level from the RM phase. Then the PEEP is reduced in 2 cmH_2O increments, each maintained for 30 seconds, until the OL-PEEP is detected. This is identified as the value corresponding to the highest dynamic compliance (defined as the PEEP value with highest Cdyn). (3) A second stepwise RM is performed in the same way as in step 1, and (4), from then on until extubation, the patients were ventilated in VCV mode using the baseline settings in combination with the detected OL-PEEP. If hemodynamic inestability (i.e. >50% decrease in mean arterial pressure) ensued during the recruitment phase, the maneuver was interrupted and 5-15 mg ephedrine or 0·05-0·15 mg phenylephrine were given. New RM were allowed in case of intraoperative hypoxemia defined as $SpO_2 < 92\%$.

Study variables

The preoperative data included the demographic variables [gender, age, height, actual body weight (ABW), PBW, and body mass index (BMI)], preoperative blood gas analysis [pH, partial pressure of arterial oxygen (PaO_2), $PaCO_2$, and the arterial oxygen saturation (SaO_2)], pulmonary function test [forced vital capacity (FVC), forced expiratory volume in the first second (FEV_1), FEV_1/FVC ratio, residual volume (RV), and diffusion of carbon monoxide (DLCO)], and the ARISCAT score which predicts the risk of suffering PPCs and includes age, surgical incision type, preoperative hemoglobin and SpO_2, presentation of a respiratory infection within the last month, surgery duration, and need for an emergency procedure.[2] Intraoperative data included the surgical procedure (Table 1), surgical site, ventilatory mode, maximum pressure achieved during the RM and, if not completed, the causes for stopping the maneuver. Finally, PEEP, cdyn, the plateau pressure, and DP were recorded pre- and post RM. Cdyn was calculated as the VT/(peak inspiratory airway pressure [Ppeak] minus the PEEP) and DP as the airway plateau pressure (Pplat) minus the PEEP. The pre-RM study variables were recorded at the beginning of the OLV and the post-RM study variables, recorded 10 minutes after the RM-PEEP, were set. PPCs were prospectively recorded by the research team during the first 7 days after surgery. All the case report forms were monitored by the same researcher (Natividad Pozo) who validated the accuracy of the data. Definitions for complications followed the standard criteria[20] and included: pneumonitis, atelectasis requiring bronchoscopy, acute respiratory failure, reintubation, pneumothorax, pneumonia, bronchopleural fistula, and acute respiratory distress syndrome (ARDS); prophylactic postoperative ventilation was not considered a PPC.

Statistical analysis

The data are expressed as the mean ± SD. The Kolmogorov–Smirnov test with Lilliefor correction was performed to test the normality of the distribution of the variables and the Levene test was used to test the homogeneity of the data. Variables with a normal distribution were analyzed by using independent Student t tests, and variables with a non-normal distribution were analyzed by using Kolmogorov–Smirnov non-parametric tests. The correlation between variables was analyzed by calculating the Pearson correlation coefficient. Statistical analysis was performed using the SPSS software package (v20.0, SPSS, Chicago, IL, USA) and statistical significance was set at $P < 0.05$.

Results

Of a population of 861 patients undergoing OLV, a total of 690 patients were included and their baseline characteristics and surgical procedures are described in Table 1. The patients were mainly male and half of them had a high-risk of PPCs (ARISCAT score exceeding 44); 11% of the patients developed at least one PPC during the first week after surgery. The occurrence of individual PPCs is described in Table 2, which also shows the patients stratified by the number of complications they presented. Table 3 shows the intraoperative variables: the mean VT was 7.6 ± 1.1 and 6.1 ± 2.1 ml/kg of the PBW during two- and OLV, respectively ($p < 0.0001$, 95% CI: 1.36–1.91). VT was identical during the pre- and post RM lung conditions; in 7.1% of patients (49 out of 690) the RM was stopped due to transient hemodynamic compromise but was successfully completed after stabilization in all cases. No other adverse events were recorded during the alveolar RMs. None of the patients required new RM due to intraoperative hypoxemia. The mean OL-PEEP significantly increased from baseline (6 cmH_2O) to reach 8.5 cmH_2O after the RM (inter-quartile range: 7–10; $p < 0.0001$, 95% CI: 2.32–2.72). DP was significantly decreased (Figure 1a) from 14 to 11 cmH_2O ($p < 0.0001$; 95% CI: −3.18 to −3.93) with the OL-PEEP, which ranged from 2 to 16 cmH_2O. Similarly, the Cdyn increased from 30.5 to 43.9 ml/cmH_2O ($p < 0.0001$; 95% CI: 12.7–14.1). With the OLA, the DP decreased in 592 patients (85.8%), remained the same in 42 (6.1%), and increased in 56 (8.1%) (Figure 1c). Neither the OL-PEEP nor the change in PEEP (ΔPEEP, calculated as the OL-PEEP minus the pre-RM PEEP) correlated with the change in DP (ΔDP, calculated as the DP after the RM minus the DP before the RM; $p = 0.232$ and 0.07, respectively). Figure 1b shows the ΔDP; we observed no correlation between the DP, plateau pressure, PEEP, and the development of PPCs ($p = 0.152$, $p = 0.748$, and $p= 0.508$, respectively).

An exploratory analysis was performed to describe the possible associations between the individualized PEEP and preoperative or intraoperative variables. There were significant—but not strong—correlations between OL-PEEP and ABW ($p < 0.001$), ideal body weight (IBW; $p = 0.005$), BMI ($p < 0.001$), FEV_1 ($p = 0.006$), and FVC ($p = 0.008$; Figure 2; Pearson correlation coefficients = 0.296, 0.110, 0.279, −0.117, and −0.118, respectively). We did not find a significant correlation between the OL-PEEP and age, height, FEV_1/FVC, RV, preoperative PaO_2, $PaCO_2$, or SaO_2. A similar pattern was found for ΔDP, which was significantly ($p < 0.001$) but not strongly correlated with ABW and BMI (Pearson correlation coefficients = 0.159 and 0.16, respectively), and no correlations were found for IBW, age, height, FEV1, CVF, FEV_1/FVC, RV, PaO_2, $PaCO_2$, or SaO_2.

Discussion

In this prospective, multicenter, national study we found an 11% incidence of PPCs in patients undergoing OLV, who received treatment with an OLA consisting of maintaining a low VT and an alveolar RM followed by an individualized PEEP to obtain best OL-PEEP respiratory system Cdyn. There was no correlation between the individualized PEEP and preoperative or surgical recorded variables and this approach significantly reduced the DP compared to pre-recruitment lung condition.

The descriptive characteristics of this study make it impossible to demonstrate a decrease in postoperative morbidity in these patients. However, our results might be of clinical relevance because there was a lower incidence (11%) of PPCs in these patients compared to previously published results for thoracic surgery: Blank et al. reporting of PPC components similar to ours showed an incidence of 18%[5] and Naik et al. found an incidence of 32%.[3,5] None of the aforementioned studies performed RMs, although they did use low PEEP (not exceeding 5 cmH_2O). To the best of our knowledge, only Licker et al., in a retrospective study, has previously applied an OLA in thoracic surgery to investigate its impact on PPCs.[4] That observational study compared OLA to conventional ventilation in an historical group of patients, and unlike our study, the PEEP level was fixed during the OLV and the RMs were performed by vital capacity maneuvers at 30-minute intervals. However, like our findings, the OLA reported by Licker et al. resulted in a reduction in PPCs when compared to the conventional ventilation approach.[4] We found an incidence of ARDS of less than 1%, which is lower than in other studies (1.2%, 2.2%, and 2.4%, respectively).[5,21,22] Again, only Licker et al., in the OLA group showed an incidence of PPS as low as ours, as well as the same incidence of atelectasis (5%)[4], which is lower than the 9% found in his conventional ventilation group or the 10% shown by Blank et al.[5] Therefore, our findings justify the implementation of a further randomized controlled trial which should be designed to compare the OLA to the commonly used protective ventilation with low VT and low PEEP levels.

The application of PEEP during OLV has become part of standard-of-care[23,24] because PEEP applied to the ventilated lung improves gas-exchange and respiratory mechanics.[10,11,25,26] Moreover, PEEP stabilizes the alveoli and mitigates atelectrauma, which might enhance the effects of protective ventilation during OLV.[27] A recent retrospective study found that low VTs without adequate PEEP did not prevent the occurrence of PPCs in patients undergoing thoracic surgery[5]. However, the impact of standard or individualized PEEP with or without RM during OLV on postoperative morbidity remains unclear.[5,7] In our opinion, several physio-pathological factors could justify the use of individualized PEEP in terms of protectiveness. Firstly, DP was decreased by RM and the individualized PEEP. Secondly, despite the concerns about the individualized PEEP as being optimal[28], it is important to highlight that by performing a decremental PEEP titration trial with a constant VT, the PEEP required for the best Cdyn indicates the lowest DP for the given VT. Thirdly, it is known that non-suitable PEEP levels may have counterproductive effects as they could promote overdistension or may not prevent alveolar re-collapse when using too high or low levels, respectively.[29,30] Regarding PEEP levels, in line with the results of our previous study[10], we found that individualized PEEPs widely varied, ranging from 2 to 16 cmH_2O. We also found 'higher' mean PEEP values in this present study compared to previous studies. This suggests that the 5 cmH_2O PEEP values which are usually recommended may not suitably provide lung protection in a significant number of patients.[24,31,32] Fourthly, the absence of strong and clinically significant association between the individualized PEEP values and the patient characteristics (such as sex, height, ABW, IBW, or BMI), patient lung function (such as FEV_1, FEV_1/FVC, PaO_2, or $SaO_{2)}$), or surgical variables (such as the surgical site) did not allow us to establish an 'optimal' and pragmatic fixed PEEP level during the OLV. Finally, as shown in this and in previous studies, the stepwise increase RM is a safe maneuver with a non-severe, transient and reversible hemodynamic compromise, but without severe consequences.[12,13] These results reinforce the hypothesis that PEEP should be individualized for each patient after a RM.

This present study is novel because it prospectively evaluates DP during the OLA for first time during OLV; DP is a surrogate of alveolar strain which is associated with inflammation and lung injury and thus, potential cause of PPCs[33]. Intraoperatively, DP seems to be associated with PPCs during two- and OLV.[5,15,16] Therefore, protective intraoperative mechanical ventilation should aim to reduce the DP and consequently, the dynamic alveolar strain. Our findings support previous studies using a strategy of RM plus PEEP during OLV.[13,14,34-36] Although the DP is not specifically given in any of these studies, we have observed in their results that the gradient of plateau pressure minus PEEP was reduced, which means a reduction in DP, even when a fixed PEEP was used. However, because many previous studies focused on gas-exchange or respiratory mechanics, they did not consider the potential protective effects of a reduction in DP. In addition, we have previously shown that after a RM, performing individualized PEEP titrations to reach the best Cdyn levels produces a greater decrease in DP than using a fixed PEEP of 5 cmH_2O with or without performing a RM.[11-13] Thus, this present study confirms our previous findings and in this respect, suggests that an OLA might be able to protect the lungs and thus, to decrease PPCs when compared to other strategy.[5]

There are several limitations in our study that need to be discussed. Due to the natural characteristics of our study, and with the absence of a control group, we were unable to establish an association between the OLA and PPCs. Therefore, the clinical interpretation of the findings must be taken with caution. Moreover, potential independent risk factors of suffering PPCs such as the duration of OLV or mechanical ventilation, the duration of surgery, the use of neuromuscular blockers and it´s reversal, the use of epidural or paravertebral analgesia techniques or perioperative fluid balance between others were not reported. All these missing information limits the extrapolation of the results to other populations. This also applies to the surgical technique as few prospective observational studies have found that VATS is associated with a lower incidence of PPCs.[37,38] Finally, the definition criteria for PPCs is not accurate between the studies reporting it´s incidence. This, together with the reporting of different number of PPCs and with the different grade of severity makes comparison between studies difficult.

In summary, in our population using low VTs together with individualized PEEPs after an alveolar RM the incidence of PPCs was lower than the incidence reported in the literature. This finding is supported by the significant decrease we found in the DP when taking an OLA. Finally, we were unable to find a pragmatic fixed PEEP level based on patient characteristics or surgical procedures, indicating that the PEEP level set should be individualized. All this justifies the implementation of a further randomized controlled trial to compare the OLA to the most commonly used protective strategy of low VT and low PEEP.

References

1. Licker MJ, Widikker I, Robert J, et al. Operative mortality and respiratory complications after lung resection for cancer: impact of chronic obstructive pulmonary disease and time trends. Ann Thorac Surg 2006;81:1830–7.
2. Canet J, Gallart Ll, Gomar C, et al; on behalf of the ARISCAT Group. Prediction of postoperative pulmonary complications in a population-based surgical cohort. Anesthesiology 2010; 113:1338-50.
3. Naik BI, Colquhoun DA, Shields IA, et al. Value of the oxygenation index during 1-lung ventilation for predicting respiratory complications after thoracic surgery. J Crit Care 2017; 37:80-84.
4. Licker M, Diaper J, Viliger Y, et al. Impact of intraoperative lung-protective interventions in patients undergoing lung cancer surgery. Critical Care 2009; 13:R41.

5. Blank R, Colquhoun D, Durieux M, et al. Management of one-lung ventilation. Impact of tidal volume on complications after thoracic surgery. Anesthesiology 2016; 124:1286-95.
6. Slutsky AS, Ranieri VM. Ventilator-induced lung injury. N Engl J Med 2013; 369:2126–36.
7. Serpa Neto A, Hemmes S, Barbas C, et al; PROVE Network Investigators. for the PROVE Network investigators. Protective versus conventional ventilation for surgery. A systematic review and individual patient data meta-analysis. Anesthesiology 2015; 123:66-78.
8. Yang D, Grant M, Stone A, et al. A meta-analysis of intraoperative ventilation strategies to prevent pulmonary complications. Is low tidal volume alone sufficient to protect healthy lungs? Ann Surg 2016; 263:881-887.
9. Spieth P, Güldner A, Carvalho A, et al. Open lung approach vs acute respiratory distress syndrome network ventilation in experimental acute lung injury. Br J Anaesth 2011;107: 388–97
10. Cinnella G, Grasso S, Raimondo P, et al. Physiological effects of the open lung approach in patients with early, mild, diffuse acute respiratory distress syndrome. Anesthesiology 2015; 123:1113-21
11. Ferrando C, Suarez-Sipmann F, Tusman G, et al. Open lung approach versus standard protective strategies: effects on driving pressure and ventilatory efficiency during anesthesia – a pilot, randomized controlled trial. PLoS One 2017; 12(5):e0177399.
12. Ferrando C, Soro M, Unzueta C, et al; Individualized PeRioperative Open-lung VEntilation (iPROVE) Network. Individualised perioperative open-lung approach

versus standard protective ventilation in abdominal surgery (iPROVE): a randomised controlled trial. Lancet Respir Med. 2018. [Epub ahead of print]

13. Ferrando C, Mugarra A, Gutierrez A, et al. Setting individualized positive end-expiratory pressure level with a positive end-expiratory pressure decremental trial after a recruitment maneuver improves oxygenation and lung mechanics during one-lung ventilation. Anesth Analg 2014; 118:657-65.
14. Unzueta C, Tusman G, Suarez-Sipmann F, et al. Alveolar recruitment improves ventilation during thoracic surgery: a randomized controlled trial. Br J Anaesth 2012; 108:517-24.
15. Ladha K, Vidal Melo M, McLean D, et al. Intraoperative protective mechanical ventilation and risk of postoperative respiratory complications: hospital based registry study. BMJ 2015; 351:h3646.
16. Neto AS, Hemmes SN, Barbas CS, et al; PROVE Network Investigators. PROVE Network Investigators. Association between driving pressure and development of postoperative pulmonary complications in patients undergoing mechanical ventilation for general anesthesia: a meta-analysis of individual patient data. Lancet Respir Med 2016; 4:272-80.
17. Maia L, Samary C, Oliveira M, et al. Impact of different ventilation strategies on driving pressure, mechanical power, and biological markers during open abdominal surgery in rats. Anesth Analg 2017; 125: 1364–1374.
18. Ferrando C, Belda FJ. Personalized intraoperative PEEP: a further step in protective ventilation. Minerva Anestesiologica 2017; 84:147-9.
19. Von Elm E, Altman DG, Egger M, et al; STROBE initiative. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet 2007; 370:1453-7.

20. Jammer Ib, Wickboldt N, Sander M, et al; European Society of Anaesthesiology (ESA) and the European Society of Intensive Care Medicine (ESICM); European Society of Anaesthesiology; European Society of Intensive Care Medicine. Standards for definitions and use of outcome measures for clinical effectiveness research in perioperative medicine: European Perioperative Clinical Outcome (EPCO) definitions. A statement from the ESA-ESCIM joint taskforce on perioperative outcome measures. Eur J Anaesthesiol 2015; 32:88-105.
21. Ruffini E, Parola A, Papalia E, et al. Frequency and mortality of acute lung injury and acute respiratory distress syndrome after pulmonary resection for bronchogenic carcinoma. Eur J Cardiothorac Surg 2001; 20:30-6.
22. Dulu A, Pastores SM, Park B, et al. Prevalence and mortality of acute lung injury and ARDS after lung resection. Chest 2006; 130:73-8.
23. Schilichting N, Flax K, Levine A, et al. Thoracic Anesthesia: A review of current topics and debates. Curr Anestesiol Rep 2016; e-ISSN 2167-6275.
24. Sentürk M. New concepts of the management of one-lung ventilation. Curr Opin Anesthesiol 2006; 19:1-4.
25. Tusman G, Böhm SH, Sipmann FS, et al. Lung recruitment improves the efficiency of ventilation and gas exchange during one-lung ventilation anesthesia. Anesth Analg 2004; 98:1604-9
26. Choi YS, Bae MK, Kim SH, et al. Effects of alveolar recruitment and positive end-expiratory pressure on oxygenation during one-lung ventilation in the supine position. Yonsei Med J 2015; 56:1421-7.
27. Karzai W, Schwarzkopf K. Hypoxemia during one-lung ventilation: prediction, prevention, and treatment. Anesthesiology 2009; 110:1402-11.

28. Bluth T, Pelosi P, Gama de Abreu M. The obese patient undergoing nonbariatric surgery. Curr Opin Anesthesiol 2016; 29:421-29.

29. Levin MA, McCormick PJ, Lin HM, et al. Low intraoperative tidal volume ventilation with minimal PEEP is associated with increased mortality. Br J Anaesth 2014; 113:97-108.

30. Dambrosio M, Roupie E, Mollet J-J, et al. Effects of positive end-expiratory pressure and different tidal volumes on alveolar recruitment and hyperinflation. Anesthesiology 1997; 87:4905-503.

31. Grichnik K, Shaw A. Update on one-lung ventilation: the use of continuous positive airway pressure ventilation and positive end-expiratory pressure ventilation – clinical application. Curr Opin Anesthesiol 2009; 22:23-30.

32. Della Rocca G, Coccia C. Ventilatory management of one-lung ventilation. Minerva Anestesiologica 2011; 77:534-6.

33. Grieco D, Chen L, Dres M, et al. Should we use driving pressure to set tidal volume? Curr Opin Crit Care 2017; 23:38-44.

34. Michelet P, D´Journo X-B, Roch A, et al. Protective ventilation influences systemic inflammation after esophagectomy: a randomized controlled study. Anesthesiology 2006; 105:911-19.

35. Cinnella G, Grasso S, Natale C, et al. Physiological effects of a lung-recruiting strategy applied during one-lung ventilation. Acta Anaesthesiol Scand 2008; 52:766-75.

36. Garutti I, Martinez G, Cruz P, et al. The impact of lung recruitment on hemodynamics during one-lung ventilation. J Cardiothorac Vasc Anesth 2009; 23:506-8.

37. Jeon J, Kang C, Kim H, et al. Video-assisted thoracoscopic lobectomy in non-small-cell lung cancer patients with chronic obstructive pulmonary disease is associated with lower pulmonary complications than open lobectomy: a propensity score-matched analysis. Eur J Cardio-Thorac Surg 2014; 45:640-5.

38. Agostini P, Lugg S, Adams K, et al. Postoperative pulmonary complications and rehabilitation requirements following lobectomy: a propensity score matched study of patients undergoing video-assisted thoracoscopic surgery versus thoracotomy. Interac Cardiovasc and Thorac Surg 2017; 1-7.

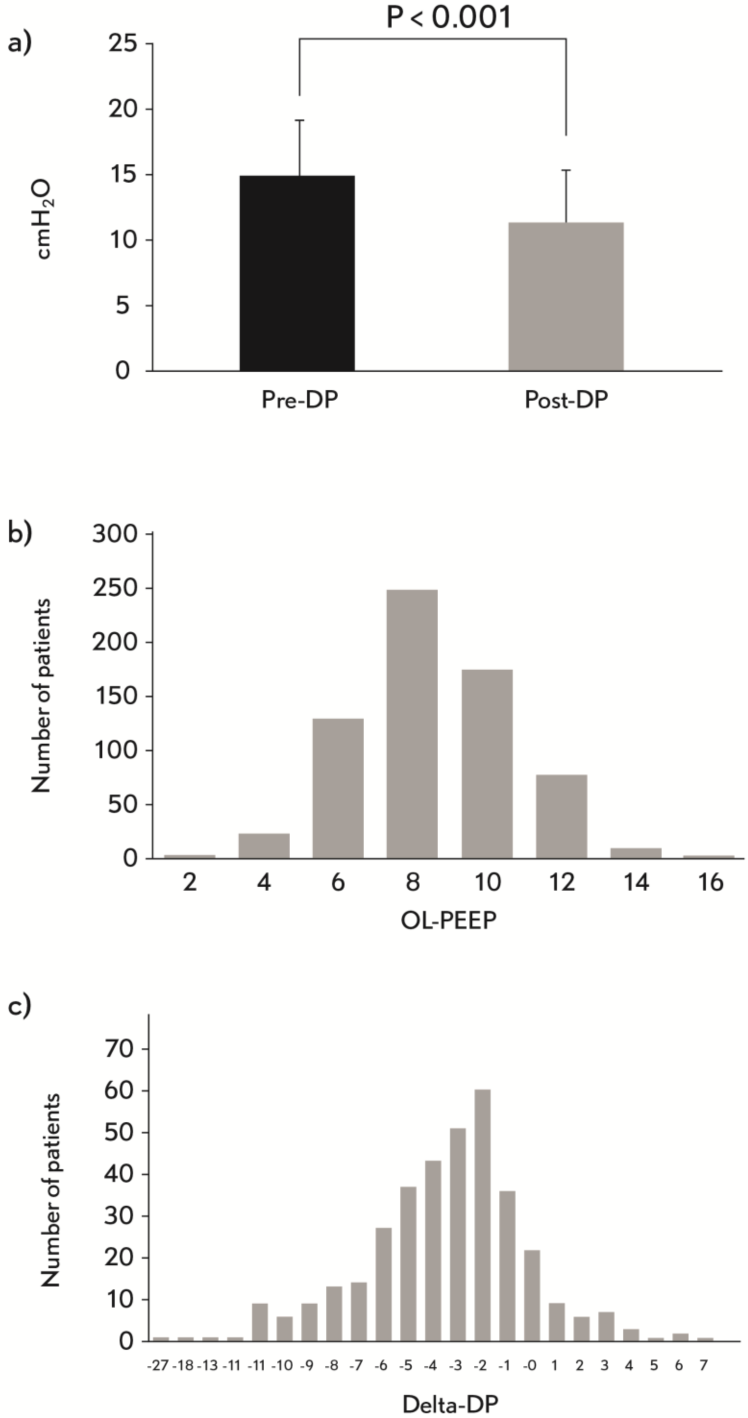
Figure legends

Figure 1.

a) OLA: open-lung approach. The driving pressure, in cmH_2O. **b)** OL-PEEP: open-lung PEEP range in cmH_2O. The OL-PEEP is the PEEP value of the best respiratory system dynamic compliance after the alveolar recruitment maneuver. **c)** Change (delta) in driving pressure, measured in cmH_2O (post-recruitment maneuver minus pre-recruitment maneuver). Driving pressure was calculated as the plateau pressure minus the PEEP.

Figure 2.

The scatterplots show the correlations between OL-PEEP: open-lung positive end-expiratory pressure, measured in cmH_2O and **a)** ABW: actual body weight in kg, **b)** IBW: ideal body weight in kg, **c)** BMI: body mass index in kg/m^2, **d)** FVC: forced vital capacity in %, **e)** FEV_1: forced expiratory volume in the first second in %. Significance and Pearson correlation coefficient: < 0.001 and 0.296; 0.005 and 0.132; < 0.001 and 0.277; 0.008 and −0.118; and 0.006 and −0.117, respectively.



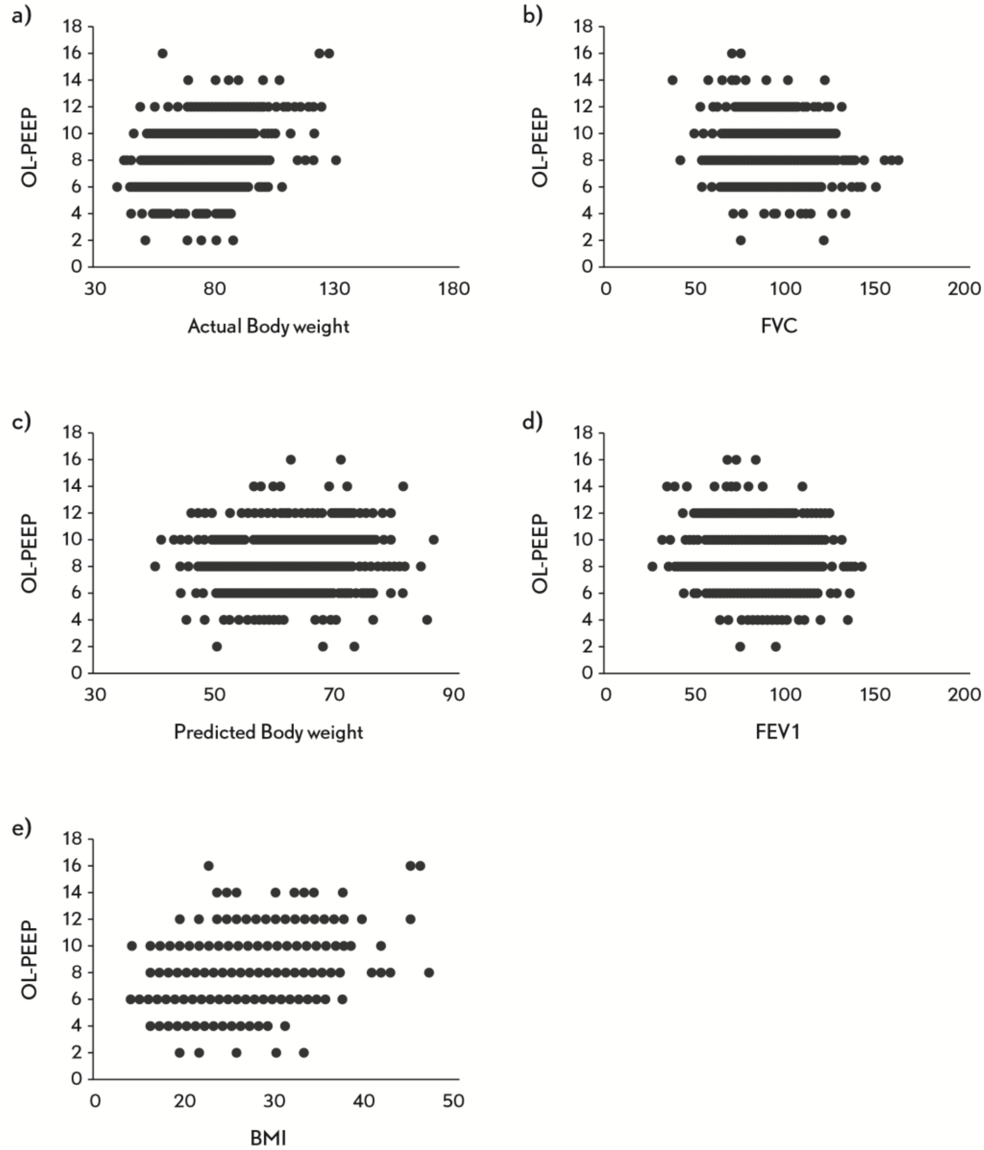

a)
OL-PEEP
18
16
14
12
10
8
6
4
2
0
30
80
130
180
Actual Body weight
b)
OL-PEEP
0
50
100
150
200
FVC
c)
OL-PEEP
30
50
70
90
Predicted Body weight
d)
OL-PEEP
0
50
100
150
200
FEV1
e)
OL-PEEP
0
20
30
40
50
BMI

Table 1. Demographic characteristics

Variable	Mean ± SD
Age	63.5 ± 11.9
Male	507 (73.4)
Height	167.2 ± 8.4
ABW	62.4 ± 8.4
PBW	75.3 ± 14.7
BMI	27.0 ± 4.8
ARISCAT	44.5 ± 12.5
FVC (%)	95.4 ± 19.8
FEV_1 (%)	85.5 ± 21.1
DLCO (%)	78.8 ± 23.1
FEV_1/FVC	77.1 ± 15.2
RV (%)	99.4 ± 40.0
PaO_2	11.4 ± 3.7
SaO_2	96.2 ± 2.0
$PaCO_2$	5.0 ± 0.6
	Number of patients
Surgical side (Right)	389 (56.2)
VATS	164 (23.6)
Wedge resection/Segmentectomy	184 (26.5)
Lobectomy	349 (50.4)
Bilobectomy/Pneumonectomy	27 (3.9)
Non-pulmonary resection	30 (4.3)
Biopsy	15 (2.1)
Other	87 (12.5)

Data are presented as mean ± standard deviation (SD) for continuous variables and n (%) for categorical variables. ABW: actual body weight in kg; PBW: predicted body weight, calculated as 50 + 0.91 (height in centimeters−152.4) for men and 45.5 + 0.91 (height in centimeters−152.4) for women; BMI: body-mass index—weight in kilograms divided by the square of the height in cm. The ARISCAT score for pulmonary complications uses three risk classes (mild, intermediate, and severe). FVC: forced vital capacity; FEV_1: forced expiratory volume in the first second; DLCO: diffusion of carbon monoxide (%); FEV_1/FVC ratio; RV: residual volume; PaO_2: partial pressure of arterial oxygen (Kpa); $PaCO_2$: partial pressure of arterial carbon dioxide (KPa); SaO_2: arterial oxyhemoglobin saturation (%); VATS: video assisted thoracoscopic surgery.

Table 2. Postoperative pulmonary complications

Incidence of PPC	**% (n)**
Atelectasis	5.3 (24)
Acute respiratory failure	6.0 (27)
Reintubation	2.7 (12)
ARDS	0.9 (4)
Pneumonia	4.4 (20)
Empyema	0.7 (3)
Bronchospasms	1.1 (5)
Pneumothorax	1.1 (5)
Pneumonitis	0 (0)
Bronchopleural fistula	0 (0)
Patients stratified by number of complications	
0	88.2 (398)
1	6.6 (30)
2	3.5 (16)
3	0.8 (4)
≥ 4	0.6 (3)

PPC: postoperative pulmonary complications; ARDS: acute respiratory distress syndrome. Only atelectasis requiring bronchoscopy was included as PPC. For the acute respiratory failure diagnosis, a SpO_2 of less than 92% was used instead of the usual 90%.

Table 3. Intraoperative ventilatory variables during one-lung ventilation

	Pre-RM	Post-RM	Variation Pre- vs. Post-RM	P-value (95 % CI)
PEEP	6 ± 1	8 ± 2	2.5 ± 2.5	p < 0.001 (2.32–2.72)
Dynamic compliance	30 ± 10	43 ± 15	13.4 ± 9.4	p < 0.001 (12.7–14.1)
Plateau pressure	20 ± 4	20 ± 4	−1.01 ± 4.14	p <0.001 (0.57–1.46)
Driving pressure	14 ± 4	11 ± 3	−3.56 ± 3.5	p < 0.001 (−3.18–−3.93)

Data are presented as the mean ± standard deviation. PEEP: positive end-expiratory pressure (cmH_2O); cdyn: dynamic compliance (ml/cmH_2O); plateau pressure (cmH_2O); the driving pressure was calculated as the plateau pressure minus the positive end-expiratory pressure (cmH_2O).